Solvency Field Theory

A Unified Note on Local Screening, Cosmic Burden–Release, and DESI DR2 Alignment

Working synthesis note

This document integrates the strongest current pieces of the program into one readable note. Its purpose is not to claim a finished fundamental theory. Its purpose is to state clearly what now appears to survive stress testing: a conservative burden/capacity grammar, a local compactness-screened Newtonian branch, a minimal bridge back to the cosmological side, and a nontrivial alignment with official DESI DR2 `base_w_wa` products in the strongest matched cases.

Status:	Working note for circulation and critique
Main posture:	No new particles, no new fields, no new dimensions
Current strongest claim:	Screened-Newton local ansatz + DESI-side phenomenological alignment
Main caution:	Not a derivation of full GR or a proof of SFT

Abstract

Solvency Field Theory (SFT) is most coherent when written as a normalized burden/capacity program rather than as a loose information metaphor. In that form, a local mass defines a compactness burden variable

$$\chi_{\text{local}}(r) = \frac{GM}{2c^2 r} = \frac{r_{\text{s}}}{4r},$$

and local gravity is written as a screened Newtonian response,

$$a(r) = a_N\, S(\chi), \qquad a_N = \frac{GM}{r^2}.$$

The preferred nonlinear branch is

$$S(\chi) = \frac{1}{1+2\chi},$$

which is simultaneously the canonically normalized member of the simplest admissible quadratic screening family, the minimizer of a minimal local screening functional, and the branch whose integrated potential is logarithmic. This branch is weak-field safe for ordinary gravitating systems, becomes relevant only for relativistic compactness, and does not reproduce strong-field Schwarzschild orbital structure. The same grammar suggests a complementary cosmological release function,

$$R(\chi) = 1 - S(\chi) = \frac{2\chi}{1+2\chi},$$

which provides a clean bridge back to the cosmological side of SFT. Finally, using official DESI DR2 `base_w_wa` products, the strongest directly comparable SFT best-fit points lie close to the extracted observationally favored w_0–w_a region, with Pantheon+ and Union3 providing the cleanest interval-level agreement. This is presented as phenomenological alignment, not validation.

1. Why this note exists

The project now has enough moving parts that a reader could miss the spine. That is the problem this note solves. The aim here is to present one coherent document that says, in order, what the framework is trying to do, what currently survives, how the local gravity extension fits the larger logic, where the observational comparison is genuinely interesting, and exactly what is and is not being claimed.

The correct tone is narrow and disciplined. This note does *not* claim a finished first-principles theory. It does claim that a conservative burden/capacity program has produced one local branch that is mathematically clean, physically interpretable, and observationally worth taking seriously.

2. The conservative workspace

The entire program is intentionally constrained to a conservative workspace:

- no new particles,
- no new fields,
- no new dimensions,
- and no appeal to ad hoc ontology simply to make the equations fit.

The working intuition is that geometry responds to normalized burden variables rather than to unconstrained metaphor. That is the unifying posture behind both the cosmological and local pieces.

3. Common burden/capacity grammar

The cleanest common language is a normalized burden variable,

$$\chi \equiv \frac{\text{burden}}{\text{capacity}}.$$

Different sectors then couple observables to different transforms of the same normalized object. Locally, the observable of interest is a screened inward response; cosmologically, the observable of interest is a released or sourced geometric response.

The preferred pair is

$$S(\chi) = \frac{1}{1+2\chi}, \qquad R(\chi) = 1 - S(\chi) = \frac{2\chi}{1+2\chi}.$$

These satisfy

$$S(0) = 1, \qquad R(0) = 0, \qquad S(\infty) = 0, \qquad R(\infty) = 1,$$

with weak-burden expansions

$$S(\chi) = 1 - 2\chi + 4\chi^2 - 8\chi^3 + \cdots,$$
$$R(\chi) = 2\chi - 4\chi^2 + 8\chi^3 - 16\chi^4 + \cdots.$$

That is already enough to show the intended duality: one branch retains response; the other releases it.

4. Local gravity extension

4.1 Compactness from burden over capacity

The local construction starts with a Bekenstein-style finite-region burden,

$$B(M, r) = 2\pi \frac{Mcr}{\hbar},$$

and a spherical area-capacity,

$$I(r) = 4\pi \frac{r^2}{\ell_*^2}.$$

Their ratio gives

$$\chi_{\text{local}}(r) = \frac{B}{I} = \frac{Mc\,\ell_*^2}{2\hbar r}.$$

With the standard area-scale identification

$$\ell_*^2 = \frac{\hbar G}{c^3},$$

one obtains

$$\chi_{\text{local}}(r) = \frac{GM}{2c^2 r} = \frac{r_{\text{s}}}{4r}, \qquad r_{\text{s}} = \frac{2GM}{c^2}.$$

This is the cleanest local result in the whole branch: normalized burden collapses naturally to a compactness variable.

4.2 Screened-force form

The main formulation should now be written as

$$a(r) = a_N\, S(\chi), \qquad a_N = \frac{GM}{r^2}.$$

In this language the Newtonian weak-field limit is immediate: any admissible branch with

$$S(0) = 1$$

recovers Newton at small compactness.

4.3 Equivalent response-potential form

The older response-potential language remains useful, but it should now be treated as equivalent rather than primary. Since

$$\frac{GM}{r} = 2c^2\chi,$$

any potential written as

$$\Phi(r) = c^2 f(\chi)$$

can be rewritten as

$$\Phi(r) = \frac{GM}{r}\, S(\chi), \qquad S(\chi) = \frac{f(\chi)}{2\chi}.$$

So the screened-force and response-potential pictures are the same theory in different coordinates. The screened-force version is simply easier to interpret physically.

5. Preferred nonlinear branch

5.1 Quadratic screening family

If one asks for the simplest admissible local variational family,

$$F(S;\chi) = \frac{a}{2}(S-1)^2 + b\,\chi S^2 + c\,\chi S + d\,\chi,$$

then enforcing weak-field admissibility and monotone screening collapses the family to

$$S(\chi) = \frac{1}{1+\lambda\chi}, \qquad \lambda = \frac{2b}{a}.$$

The canonically normalized member is

$$S(\chi) = \frac{1}{1+2\chi}.$$

So the preferred branch is not an isolated guess. It is the normalized member of the simplest admissible quadratic screening family.

5.2 Minimal variational origin

The cleanest toy origin is the minimal local screening functional

$$F(S;\chi) = \frac{1}{2}(S-1)^2 + \chi S^2.$$

Extremizing gives

$$\frac{\partial F}{\partial S} = 0 \quad \Rightarrow \quad (S-1) + 2\chi S = 0 \quad \Rightarrow \quad S(\chi) = \frac{1}{1+2\chi}.$$

So the preferred branch is also the minimizer of the simplest local quadratic screening cost balancing unscreened response against compactness-weighted penalty.

5.3 Force law and logarithmic potential

With

$$S(\chi) = \frac{1}{1+2\chi},$$

the screened force law becomes

$$a_{\log}(r) = \frac{a_N}{1+2\chi} = \frac{a_N}{1+GM/(c^2 r)}.$$

Integrating from r to infinity gives

$$\Phi_{\log}(r) = c^2 \log\left(1 + \frac{GM}{c^2 r}\right).$$

This is why the branch is best described as a *compactness-screened Newtonian ansatz*. It is not merely a convenient curve; it is the exact potential generated by a first-order compactness-screened Newtonian force law.

6. What the local branch is and is not

The local branch now has a clean classification.

Current best reading of the local branch

- It **is** a conservative screening correction to Newtonian gravity.
- It **is** weak-field safe for ordinary gravitating systems.
- It **does** become relevant for high compactness, especially neutron-star and black-hole environments.
- It **is not** a replacement for full GR.
- It **does not** reproduce ISCO-like Schwarzschild orbital structure.
- It **should not** be sold as a black-hole pseudo-potential.

That last point matters. The branch survives specifically because it is narrow and honest. It screens Newton; it does not pretend to have reproduced strong-field relativistic structure.

7. Bridge back to cosmology

The clean bridge is not to force local and cosmological equations to be literally identical. The clean bridge is to use the same normalized burden grammar in both sectors and couple observables differently.

Locally,

$$a_{\text{local}} = a_N\, S(\chi_{\text{local}}).$$

Cosmologically, the natural complementary source is

$$R(\chi) = 1 - S(\chi) = \frac{2\chi}{1+2\chi}.$$

This suggests the minimal dual coupling

$$a_{\text{local}} = a_N\, S(\chi), \qquad \mathcal{G}_{\text{cosmic}} \propto R(\chi),$$

where the local observable couples to retained response and the cosmological observable couples to released response.

That bridge is still a conjectural unification step, but it is the cleanest one available. It means the local screening branch is not an unrelated curiosity; it becomes one side of a common burden–response grammar.

8. Observational alignment with DESI DR2

The most important empirical checkpoint so far is not a proof, but it is real: the strongest SFT best-fit points land close to the official DESI DR2 `base_w_wa` landscape in the best directly comparable cases.

The cleanest rows are Pantheon+ and Union3. Using extracted DESI-side central values and intervals from the official products, the matched SFT points land very close to the DESI-side w_0–w_a coordinates, and in both of these headline rows the SFT values sit inside the extracted 68-style intervals.

Table 1: Headline SFT vs. DESI DR2 comparison rows. These are the strongest directly comparable cases and should be treated as the main phenomenological alignment table.

DESI dataset	DESI w	DESI w_a	SFT w_0	SFT w_a
Union3 + DESI BAO + Planck + lensing	-0.667 ± 0.088	$-1.090^{+0.310}_{-0.270}$	-0.663	-1.077
Pantheon+ + DESI BAO + Planck + lensing	-0.838 ± 0.055	$-0.620^{+0.220}_{-0.190}$	-0.845	-0.596

The DESY5SN-flavored rows are still supportive, but they are looser and should not be the headline examples. Rows with only one-sided DESI constraints, incomplete central values, or no direct SFT analogue should not be framed as failures. They are simply weaker comparison cases.

The correct interpretation is therefore modest but meaningful: the framework is not merely narratively compatible with the public dark-energy landscape; in the strongest matched cases it lands in the same neighborhood quantitatively.

9. Claims and nonclaims

9.1 What this note claims

- A normalized burden/capacity program gives a clean local compactness variable,

$$\chi_{\text{local}} = \frac{GM}{2c^2 r} = \frac{r_{\text{s}}}{4r}.$$

- The preferred local branch is a screened-Newton ansatz,

$$a(r) = a_N \frac{1}{1 + 2\chi}.$$

- That branch has a minimal quadratic variational origin.
- The same burden grammar suggests a complementary cosmological release function,

$$R(\chi) = \frac{2\chi}{1 + 2\chi}.$$

- In the strongest directly comparable rows, SFT best-fit points align nontrivially with official DESI DR2 w_0–w_a products.

9.2 What this note does not claim

- It does not derive Einstein's equations from first principles.
- It does not derive the value of G from first principles.
- It does not reproduce Schwarzschild or Kerr strong-field structure.
- It does not validate SFT.
- It does not turn every unmatched or limit-only observational row into evidence for or against the theory.

10. Recommended support bundle

If this note is circulated, the support package should be kept small and legible. The recommended bundle is:

- `paper_table_main.csv` - headline DESI comparison rows.
- `paper_table_appendix.csv` - broader comparison set.
- `paper_table_notes.txt` - interpretation guardrails for the comparison bundle.
- `local_gravity_note.pdf` - technical local companion if someone wants the standalone derivation path.

The main note should always be shown first. The support bundle exists to backstop, not to replace, the main document.

11. Conclusion

The important change is not that every open problem is solved. The important change is that the surviving pieces now fit together in one coherent hierarchy.

The framework is strongest when it is read as a normalized burden/capacity program. In that form, the local extension produces a conservative compactness-screened Newtonian branch with a clean variational origin. That branch does not pretend to be a full theory of relativistic gravity, but it is narrow enough to defend. The cosmological side is then most naturally interpreted through the complementary release function rather than through a disconnected phenomenological closure. Finally, the strongest directly comparable DESI DR2 rows show that the SFT best-fit points are not wandering arbitrarily through parameter space; they are landing near the public observationally favored dark-energy sector.

That is enough to justify showing people one coherent document instead of a pile of fragments.